

To do
BVP

4.1(1-22)

$$1 \quad y = c_1 e^x + c_2 e^{-x}, \quad (-\infty, \infty);$$

$$y'' - y = 0, \quad y(0) = 0, \quad y'(0) = 1$$

$$y(0) = 0 \quad 0 = c_1 e^0 + c_2 e^{-0}$$

$$0 = c_1 + c_2 \rightarrow \text{eq. (i)}$$

$$y' = c_1 e^x - c_2 e^{-x}$$

$$y'(0) = 1 \quad 1 = c_1 e^0 - c_2 e^{-0}$$

$$1 = c_1 - c_2 \rightarrow \text{eq. (ii)}$$

$$\text{eq. (i)} + \text{eq. (ii)}$$

$$c_1 + c_2 = 0$$

$$c_1 - c_2 = 1$$

$$2c_1 = 1$$

$$c_1 = 1/2$$

$$\text{Put } c_1 \text{ in eq. (i)}$$

$$1/2 + c_2 = 0$$

$$c_2 = -1/2$$

Put c_1 and c_2 in question

$$y = \frac{1}{2} e^x - \frac{1}{2} e^{-x}$$

$$4 \quad y = c_1 + c_2 \cos x + c_3 \sin x, \quad (-\infty, \infty)$$

$$y''' + y' = 0, \quad y(\pi) = 0, \quad y'(\pi) = 2$$

$$y''(\pi) = -1$$

$$y(\pi)=0 \quad 0 = C_1 + C_2 \cos \pi + C_3 \sin \pi$$

$$0 = C_1 - C_2 \rightarrow \text{eq. (i)}$$

$$y' = -C_2 \sin x + C_3 \cos x$$

$$y'(\pi) = 2 \quad 2 = -C_2 \sin \pi + C_3 \cos \pi$$

$$2 = -C_3$$

$$C_3 = -2 \rightarrow \text{eq. (ii)}$$

$$y'' = -C_2 \cos x - C_3 \sin x$$

$$y''(\pi) = -1 \quad -1 = -C_2 \cos \pi - C_3 \sin \pi$$

$$-1 = C_2 \rightarrow \text{eq. (iii)}$$

Put C_2 in eq. (i)

$$C_1 + 1 = 0$$

$$C_1 = -1$$

Put in eq.

$$y = -1 + \cos x - 2 \sin x$$

$$7 \quad x(t) = C_1 \cos \omega t + C_2 \sin \omega t$$

$$x'' + \omega^2 x = 0$$

$$x(0) = x_0, \quad x'(0) = x_1$$

$$x(t) = x_0 \cos \omega t + \frac{x_1}{\omega} \sin \omega t$$

$$x'(t) = -C_1 \omega \sin \omega t + C_2 \omega \cos \omega t$$

$$x(0) = x_0 \quad x_0 = C_1 \cos \omega(0) + C_2 \sin \omega(0)$$

$$x_0 = C_1$$

$$x'(0) = 1 \quad \mu = c_1 \omega$$

$$c_2 = \frac{\mu}{\omega}$$

Put in eq.

$$x(t) = x_0 \cos \omega t + \frac{\mu}{\omega} \sin \omega t$$

$$10 \quad y'' + (\tan x)y = e^x$$

$$y(0) = 1, \quad y'(0) = 0$$

$$\tan x = 0$$

$$(-\pi/2, \pi/2)$$

$$11 \quad y'' - y = 0$$

$$y(0) = 0, \quad y(1) = 1$$

$$a \quad y = c_1 e^x + c_2 e^{-x}$$

$$0 = c_1 + c_2 \rightarrow \text{eq. (i)}$$

$$1 = c_1 e + c_2 e^{-1} \rightarrow \text{eq. (ii)}$$

x eq. (i) by e

$$c_1 e + c_2 e = 0$$

$$+ c_1 e + c_2 e^{-1} = 1$$

$$c_2(e - e^{-1}) = -1$$

$$c_2 = \frac{-1}{e - e^{-1}}$$

$$= \frac{-e}{e^2 - 1}$$

Put c_2 in eq. (1)

$$c_1 + c_2 = 0$$

$$c_1 - \frac{e}{e^2 - 1} = 0$$

$$\therefore c_1 = \frac{e}{e^2 - 1}$$

$$y = \frac{e}{e^2 - 1} e^{e^2 - 1} - \frac{e}{e^2 - 1} e^{-x}$$

$$b \quad y = c_3 \cosh x + c_4 \sinh x$$

$$y(0) = 0 \quad 0 = c_3 \cosh(0) + c_4 \sinh(0)$$

$$c_3 = 0$$

$$y(1) = 1 \quad 1 = c_3 \cosh 1 + c_4 \sinh 1$$

Not possible

c No they are not equivalent

$$15 \quad \text{L.I. } f_1(x) = x, \quad f_2(x) = x^2$$

$$f_3(x) = 4x + 3x^2$$

$W \neq 0$
 $W = 0$ L.D

$$W = \begin{vmatrix} f_1 & f_2 & f_3 \\ f_1' & f_2' & f_3' \\ f_1'' & f_2'' & f_3'' \end{vmatrix} = \begin{vmatrix} x & x^2 & 4x + 3x^2 \\ 1 & 2x & 4 + 6x \\ 0 & 2 & -6 \end{vmatrix}$$

$$\begin{aligned}
 &= x(-12x - 8 + 12x) - x^2(-6) \\
 &\quad + (4x - 3x^2)(2) \\
 &= -8x + 6x^2 + 8x - 6x^2 \\
 &= 0
 \end{aligned}$$

$W = 0$ So it is L.D

20 $f_1(x) = 2 + x$, $f_2(x) = 2 + |x|$

$$W = \begin{vmatrix} f_1 & f_2 \\ f_1' & f_2' \end{vmatrix}$$

$$= \begin{vmatrix} 2+x & 2+|x| \\ 1 & 1 \end{vmatrix}$$

$$\begin{aligned}
 &= 2+x - 2+x \\
 &= 0
 \end{aligned}$$

$W \neq 0$

So it is L.I

21 $f_1(x) = 1 + x$, $f_2(x) = x$, $f_3(x) = x^2$

$$W = \begin{vmatrix} 1+x & x & x^2 \\ 1 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix}$$

$$= (1+x)(2) - x(2) + x^2(0) = 2 + 2x - 2x$$

$$W \neq 0$$

So it is L.I

Ex 4.2 (1-20)

Reduction of order

There are 2 methods.

order = no. of sol

Method 1:

$$y_2 = u y_1$$

Then take the derivatives as required then take integration to find u .

Method 2:

$$y_2 = y_1 \int \frac{e^{-\int p(x) dx}}{y_1^2} dx$$

If the eq. is in form

$$y'' + p(x)y' + q(x)y = 0$$

$$1 \quad y'' - 4y' + 4y = 0; \quad y_1 = e^{2x}$$

$$\begin{aligned} y_2 &= e^{2x} \int \frac{e^{-\int 4 dx}}{(e^{2x})^2} dx \\ &= e^{2x} \int \frac{e^{-4x}}{e^{4x}} dx \end{aligned}$$

$$y_2 = ne^{2x} + c$$

$$y_2 = ne^{2x} \quad \because c=0$$

4 $y'' + 9y = 0$; $y_1 = \sin 3x$

$$y_2 = \sin 3x \int \frac{e^{\int 0 dx}}{(\sin 3x)^2} dx$$

$$= \sin 3x \int \frac{e^0}{(\sin 3x)^2} dx$$

$$= \sin 3x \int (\operatorname{cosec} 3x)^2 dx$$

$$= -\sin 3x \cot 3x + c$$

$$= -\sin 3x \cdot \frac{\cos 3x}{\sin 3x} + c$$

$$= -\cos 3x + \frac{\sin 3x}{1}$$

$$= -\cos 3x \quad \because c=0$$

7 $9y'' - 12y' + 4y = 0$ $y_1 = e^{2x/3}$

$$y'' - \frac{4}{3}y' + \frac{4}{9}y = 0$$

$$y_2 = e^{2x/3} \int \frac{e^{\int 4/3 dx}}{(e^{2x/3})^2} dx$$

$$= e^{2x/3} \int \frac{e^{4/3 x}}{e^{4/3 x}} dx$$

$$= xe^{2x/3} + c$$

$$y_2 = xe^{2x/3} \quad \because c=0$$

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$$x^2 y'' - xy' + 2y = 0$$

$$y'' - \frac{1}{x} y' + \frac{2y}{x^2} = 0 \quad y_1 = x \sin(\ln x)$$

$$y_2 = x \sin(\ln x) \int \frac{e^{\int x dx}}{x^2 \sin^2(\ln x)} dx$$

$$= x \sin(\ln x) \int \frac{e^{x^2/2}}{x^2 \sin^2(\ln x)} dx$$

$$= x \sin(\ln x) \int \frac{e^{x^2/2}}{x^2} \frac{\operatorname{cosec}^2(\ln x)}{x^2} dx$$

ILATE

$$= x \sin(\ln x) \frac{\operatorname{cosec}^2(\ln x)}{x^2} \frac{e^{x^2/2}}{x} - \int$$

$$\frac{e^{x^2/2}}{x} \cdot x^{-2} \cdot 2 \operatorname{cosec} \ln x$$

$$y_2 = x \sin(\ln x) \int \frac{e^{\int 1/x dx}}{x^2 \sin^2 \ln x} dx$$

$$= x \sin(\ln x) \int \frac{e^{\ln x}}{x^2 \sin^2 \ln x} dx$$

$$= x \sin(\ln x) \int \frac{\operatorname{cosec}^2 \ln x}{x} dx$$

$$= -x \sin(\ln x) \cot(\ln x) + C$$

$$= -x \cos(\ln x) \quad \because C=0$$

$$20 \quad y'' - 4y' + 3y = x \quad ; \quad y_1 = e^x$$

$$y_2 = u y_1 = u e^x$$

$$y_1' = u e^x + e^x \frac{du}{dx}$$

$$y_2'' = u e^x + e^x \frac{du}{dx} + e^x u'' + u' e^x$$

$$= u e^x + 2u' e^x + u'' e^x$$

Put

$$u e^x + 2u' e^x + u'' e^x - 4u e^x - 4u' e^x + 3u e^x = x$$

$$-2u' e^x + u'' e^x = x$$

$$e^x (u'' - 2u') = x$$

$$u'' - 2u' = 0$$

$$I.F = e^{-2x}$$

$$e^{-2x} u'' - 2e^{-2x} u' = 0$$

$$\int (e^{-2x} \cdot u')' dx = \int 0 dx$$

$$(e^{-2x} \cdot u') = C$$

$$\int u' dx = \int C e^{2x} dx$$

$$u = C \frac{e^{2x}}{2}$$

$$u = C e^{2x} / 2$$

$$y_2 = u e^{2x} \cdot e^x$$

$$= C e^{3x} \quad \therefore C = 0$$

$$y_2 = e^{3x}$$

homogeneous

Ex 4.3 (1-40)

real and distinct

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

real and equal

$$y_c = c_1 e^{mx} + c_2 x e^{mx}$$

imaginary

$$y_c = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$$

$$1 \quad 4y'' + y' = 0$$

$$4m^2 + m = 0$$

$$m(4m+1) = 0$$

$$m_1 = 0$$

$$m_2 = -1/4$$

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$= c_1 + c_2 e^{-1/4 x}$$

$$1 \quad 12y'' - 5y' - 2y = 0$$

$$12m^2 - 5m - 2 = 0$$

$$12m^2 + 3m + 8m - 2 = 0$$

$$3m(4m+1) - 2(4m+1) = 0$$

$$(3m-2)(4m+1) = 0$$

$$m_1 = \frac{2}{3}, \quad m_2 = -\frac{1}{4}$$

$$y = c_1 e^{2/3 x} + c_2 e^{-1/4 x}$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$19 \quad \frac{d^3 u}{dt^3} + \frac{d^2 u}{dt^2} - 2u = 0$$

$$m^3 + m^2 - 2 = 0$$

$$\begin{array}{c|cccc} 1 & 1 & 1 & 0 & -2 \\ & & 1 & 2 & 2 \\ \hline & & 1 & 2 & 2 & 0 \end{array}$$

$$m^2 + 2m + 2 = 0$$

$$m = \frac{-2 \pm \sqrt{4 - 4(1)(2)}}{2(1)}$$

$$= \frac{-2 \pm \sqrt{4 - 8}}{2} = \frac{-2 \pm 2i}{2}$$

$$= -1 \pm i$$

$$m_1 = 1 \quad m_2 = -1 \pm i$$

$$y = c_1 e^x + e^{-x} (c_2 \cos x + c_3 \sin x)$$

$$23 \quad y^{(4)} + y''' + y'' = 0$$

$$m^4 + m^3 + m^2 = 0$$

$$m^2 (m^2 + m + 1) = 0$$

$$m^2 = 0 \quad m^2 + m + 1 = 0$$

$$m = 0 \quad m = \frac{-1 \pm \sqrt{1 - 4}}{2}$$

$$m = 0$$

$$2$$

$$= \frac{-1 \pm \sqrt{3}i}{2}$$

$$2$$

$$y = c_1 + e^{-\frac{1}{2}x} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right)$$

$$27 \quad \frac{d^5 u}{dr^5} + 5 \frac{d^4 u}{dr^4} - 2 \frac{d^3 u}{dr^3} - 10 \frac{d^2 u}{dr^2} + \frac{du}{dr} + 5u = 0$$

$$m^5 + 5m^4 - 2m^3 - 10m^2 + m + 5 = 0$$

$$\begin{array}{c|cccccc} 1 & 1 & 5 & -2 & -10 & 1 & 5 \\ & & 1 & 6 & 4 & -6 & -5 \\ \hline & 1 & 6 & 4 & -6 & -5 & 0 \end{array}$$

$$m_1 = 1$$

$$m^4 + 6m^3 + 4m^2 - 6m - 5 = 0$$

$$\begin{array}{c|cccccc} 1 & 1 & 6 & 4 & -6 & -5 \\ & & 1 & 7 & 11 & 5 \\ \hline & 1 & 7 & 11 & 5 & 0 \end{array}$$

$$m_2 = 1$$

$$m^3 + 7m^2 + 11m + 5 = 0$$

$$\begin{array}{c|cccc} -1 & 1 & 7 & 11 & 5 \\ & & -1 & -6 & -5 \\ \hline & 1 & 6 & 5 & 0 \end{array}$$

$$m^2 + 6m + 5 = 0$$

$$\begin{array}{c|ccc} -1 & 1 & 6 & 5 \\ & & -1 & -5 \\ \hline & 1 & 5 & 0 \end{array}$$

$$m + 5 = 0$$

$$m = -5$$

$$36 \quad y''' + 2y'' - 5y' - 6y = 0$$

$$y(0) = y'(0) = 0 \quad y''(0) = 1$$

$$m^3 + 2m^2 - 5m - 6 = 0$$

$$\begin{array}{c|cccc} -1 & 1 & 2 & -5 & -6 \\ & & -1 & -1 & 6 \\ & 1 & 1 & -6 & 0 \end{array}$$

$$m_1 = -1$$

$$m^2 + m - 6 = 0$$

$$m^2 + 3m - 2m - 6 = 0$$

$$m(m+3) - 2(m+3) = 0$$

$$(m-2)(m+3) = 0$$

$$m_2 = -3$$

$$m_3 = 2$$

$$y = c_1 e^{-x} + c_2 e^{-3x} + c_3 e^{2x}$$

$$y(0) = 0 \quad 0 = c_1 + c_2 + c_3 \quad \begin{array}{l} c_1 + 5/2 c_3 = 0 \\ c_1 = -5/2 c_3 \end{array}$$

$$y' = -c_1 e^{-x} - 3c_2 e^{-3x} + 2c_3 e^{2x}$$

$$y'(0) = 0 \quad 0 = -c_1 - 3c_2 + 2c_3$$

$$y'' = c_1 e^{-x} + 9c_2 e^{-3x} + 4c_3 e^{2x}$$

$$y''(0) = 1 \quad 1 = c_1 + 9c_2 + 4c_3$$

$$c_1 + c_2 + c_3 = 0$$

$$-c_1 - 3c_2 + 2c_3 = 0$$

$$-2c_2 + 3c_3 = 0$$

$$c_2 = \frac{-3c_3}{-2} = \frac{3c_3}{2}$$

$$\frac{-5}{2} c_3 + \frac{9 \times 3}{2} c_3 + 4 c_3 = 1$$

$$\frac{-5 + 27 + 8}{2} c_3 = 1$$

$$\frac{30}{2} c_3 = 1$$

$$c_3 = 1/15$$

Put c_3 in eq. (i) (ii)

$$c_1 + c_2 = -1/15$$

$$-1 - 3c_2 = -2\left(\frac{1}{15}\right)$$

$$-2c_2 = \frac{-1}{15} - \frac{2}{15}$$

$$-2c_2 = \frac{-1-2}{15}$$

$$c_2 = \frac{-3}{15 \times -2}$$

$$c_2 = \frac{1}{10}$$

Put c_3 and c_2 in eq. (i)

$$c_1 + \frac{1}{10} + \frac{1}{15} = 0$$

$$c_1 = \frac{-1}{10} - \frac{1}{15}$$

$$= \frac{-3-2}{30} = \frac{-5}{30} = \frac{-1}{6}$$

$$\text{So } y = \frac{-1}{6} e^{-x} + \frac{1}{10} e^{-3x} + \frac{1}{15} e^{2x}$$

40 $y'' - 2y' + 2y = 0$ $y(0) = 1, y(\pi) = 1$

$$m^2 - 2m + 2 = 0$$

$$m = \frac{2 \pm \sqrt{4 - 4(1)(2)}}{2(1)}$$

$$= \frac{2 \pm 2i}{2} = 1 \pm i$$

$$y = e^x (C_1 \cos x + C_2 \sin x)$$

$$y(0) = 1 \quad 1 = C_1$$

$$y(\pi) = 1 \quad 1 = -e^\pi$$

$$-e^\pi \neq 1$$

So it has no sol.

Ex 4.4 [UC]

for non-homogenous

$g(x) = \text{constant, polynomial, } e^{ax}, \sin x, \cos x$
or multiple of any of it

18 $y'' - 2y' + 2y = e^{2x} (\cos x - 3 \sin x)$

$$m^2 - 2m + 2 = 0$$

$$m = \frac{2 \pm \sqrt{4 - 4(1)(2)}}{2}$$

$$= \frac{2 \pm 2i}{2} = 1 \pm i$$

$$y_c = e^x (C_1 \cos x + C_2 \sin x)$$

$$y_p = e^{2x} (A \cos x + B \sin x)$$